Unifying Benchmark Evaluation Protocols: A Meta-Evaluation Standard for Artificial Intelligence in Requirements Engineering

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Abstract. Context and motivation: As foundation models accelerate software implementation, the engineering bottleneck shifts upstream to validating whether candidate models satisfy system requirements. Teams use public benchmark scores and automated reviews as acceptance evidence, so engineers must verify whether a benchmark and its scoring harness support the requirement.
Question/problem: Published benchmark review frameworks ask different questions and apply conflicting rules when documentation is silent, reaching contradictory verdicts on the same benchmark without distinguishing located code defects from unreported procedures.
Principal ideas/results: We unify seven review frameworks into MESA, mapping 280 items (386 links) into 30 concerns (C01–C30) and deriving a 39-criterion rubric (8 descriptive A01–A08; 31 evaluative B01–B29) recording evidence level, check outcome, check kind, scope, detection limits, and task defects. Across 18 sessions applying six frameworks to four benchmarks (HumanEval, TruthfulQA, HLE, τ-bench), frameworks agreed on 159/302 comparisons (53%); 92/143 disagreements (64%) stemmed from item granularity (50) and missing-documentation rules (42), with 210/711 answers carrying the *unreported* qualifier versus 13 contradictions. Applied twice per benchmark, MESA matched on 26/32 Part A classifications (81%, $\alpha_n = 0.80$), 99/124 Part B levels (80%, $\alpha_o = 0.70$), and 83/124 level-and-outcome pairs (67%). Auditing all 312 judgments confirmed 55/64 Part A and 159/248 Part B ratings directly, with 54 plausible absences, 29 weaker-support ratings, 6 contradicted absences, and 0 fabricated quotes.
Contribution: (1) A 280-item mapping and 39-criterion rubric separating absent reporting from located defects; (2) an empirical analysis of framework disagreements and rubric repeatability; and (3) a Requirements Traceability Matrix design gating acceptance decisions.

Keywords: Requirements Engineering · Artificial Intelligence · Benchmarks · Acceptance Testing · Construct Validity · Traceability.

1 Introduction

As foundation models and coding assistants accelerate software implementation, the engineering bottleneck shifts upstream to validating whether candidate models satisfy system requirements [11]. To keep pace with rapid model releases, engineering teams rely on public benchmark scores and automated reviews as acceptance evidence for

capabilities such as code generation, question answering, or tool use. A leaderboard score, however, does not establish that a model satisfies a system requirement [24,2]. Consider HumanEval [21], a widely used Python code synthesis benchmark: its paper describes executing generated code inside a gVisor container sandbox, yet its released execution script runs untrusted candidate completions via `exec()` in the same Python process and namespace as the test assertions under a comment stating that the guard is not a security sandbox. Simultaneously, the HumanEval paper does not report whether the scoring harness was tested against adversarial output exploits. An acceptance process that compresses validation into a single score risks either admitting contaminated or unisolated benchmark evidence or rejecting a candidate model because a benchmark paper omitted a reporting detail.

To audit benchmarks as measurement instruments, researchers have introduced review checklists and evaluation protocols [23,19,14,5,28,1,6]. While recent guidelines such as NIST AI 800-2 [20] and machine learning quality models [17] call for aligning benchmark selection with system goals, each review framework was designed independently with its own item scope, rating scale, and rule for absent documentation.

When applied to the same benchmark snapshot, published review frameworks frequently reach opposite conclusions. On HumanEval's scoring validation (Concern C07), for example, BetterBench awards its maximum score on its single validation item, whereas Scorecard marks four of its eight scoring items unsatisfied because adversarial parser checks are not reported in the paper. Across 18 review sessions on four benchmarks in our study, frameworks agreed on 159 of 302 pairwise comparisons (53%). Across the 711 recorded item answers, 210 answers (30%) penalized unreported information as a failure (pooling unmet reporting requirements with unverified practices), whereas 13 answer instances recorded confirmed contradictions with the benchmark artifacts. On the HumanEval example above, a framework that scores unreported exploit testing as 0 and a review that locates the unisolated `exec()` call in the execution script both produce negative scores, even though the first reflects absent documentation (*Undocumented*, check outcome *not performed*, triggering **Inquiry Required**) and the second reflects a located defect (*Contradicted* or *Verified* with check outcome *failed*, triggering **Inadmissible**).

This paper joins seven published benchmark evaluation protocols into a unified meta-evaluation standard called MESA (Meta-Evaluation Standard for AI), grounded in construct validity [15,3], representation theory [25], and the EFPA test review model [8]: (1) **Seven-Framework Mapping and Rubric:** mapping 280 published items across 386 links into 30 shared concerns (C01–C30) and deriving a 39-criterion rubric (8 Part A descriptive, 31 Part B evaluative) recording evidence levels, check outcomes, check kinds, scopes, detection limits, and defects; (2) **Three-Tier Verdict Architecture:** separating Tier 1 rubric levels, Tier 2 harmonized framework verdicts (**Satisfied**, **Unsatisfied**, **Open**) qualified by absent reporting ($-r$), and Tier 3 requirements acceptance statuses; (3) **Requirements Traceability Matrix Design:** linking system capability requirements to audited benchmark evidence, specifying how evidence gates admissibility before score thresholding; and (4) **Empirical Analysis of Disagreements and Repeatability:** evaluating pairwise agreement, coding 143 conflicts, and measuring repeatability across four benchmark snapshots (HumanEval, TruthfulQA, HLE, τ-bench).

2 Background and Motivation

2.1 Using Benchmark Scores as Evidence for Requirements

In classical Requirements Engineering, the Jackson-Zave reference model defines satisfaction as $W, S \vdash R$, where domain assumptions W and software specification S deductively entail system requirements R [27,11]. When integrating a foundation model M, engineers do not have an inspectable specification S of the model's weights that can be proven to satisfy a capability requirement G_{AI}. Instead, teams argue from test evidence: if model M achieves at least a target score θ on a benchmark B executed in environment E under operational assumptions W, that observation supports the claim that M satisfies G_{AI}. Formally:

$$W, E, O_{\mathrm{bench}}(M, B) \vdash_{\mathrm{ind}} G_{\mathrm{AI}} \tag{1}$$

where $B = (T, H, S_{\mathrm{score}})$ is the benchmark instrument comprising task suite T, execution harness H, and scoring rule S_{score} [4], and $\vdash_{\mathrm{ind}}$ denotes an inductive assurance argument [12].

In assurance frameworks such as AMLAS, ISO/IEC 25059, and machine learning component quality models [12,13,17], the strength of this argument depends on whether benchmark B is a trustworthy measurement instrument for G_{AI}. The argument fails in two classic ways: (1) **Construct underrepresentation (incomplete coverage)**, where tasks T test narrow syntax rather than the capability demanded by G_{AI} (e.g., single-function docstring completion when G_{AI} requires repository bug fixing); and (2) **Construct-irrelevant variance (score distortion)**, where scores are inflated by test-runner exploits in H (such as unisolated `exec()` calls), weak assertions in S_{score}, or pretraining exposure of T.

2.2 Construct Validity as an Assurance Argument: Operationalizing Kane

In educational and psychological measurement, validity is the degree to which evidence supports a proposed interpretation and use of test scores [15,3]. Under Kane's argument-based framework, using a benchmark score to support requirements decisions spans four inferential stages: (1) **Scoring Inference ($I_{\mathbf{score}}$):** Was each task scored accurately? Evaluates whether scoring rules are well defined and test runners cannot be gamed (Criteria B06, B07). (2) **Generalization Inference ($I_{\mathbf{gen}}$):** Does the observed score hold across tasks, prompts, and runs? Evaluates sampling, baselines, prompt stability, repeatability, and uncertainty (B08–B12, B20). (3) **Extrapolation Inference ($I_{\mathbf{extrap}}$):** Do tasks represent the intended capability without shortcuts or exposure? Evaluates construct definition, task realism, reference answers, shortcuts, exposure, and baselines (B01, B03–B05, B13–B16, B23, B27). (4) **Decision Inference ($I_{\mathbf{dec}}$):** What operational decision does the evidence support? Evaluates stated use, documentation, replication, licensing, maintenance, safety, and isolation (B02, B17–B19, B21–B22, B24–B26, B28a/b, B29). Following the EFPA test review model [8], MESA separates neutral descriptive benchmark properties (Part A) from evaluative criteria across these four stages (Part B).

Table 1. The seven review frameworks and the unified comparison standard.

Code	Framework	Paper	Items	Scope	Rule for Missing Documentation	Standardized Verdict
BB	BetterBench [23]	46	47	General AI	Score 0 (unacknowledged); Score 5 (no proof)	Unsatisfied, *unreported*
ECBD	ECBD [19]	20	20	NLP	Absence of support recorded without score	Open
SC	Scorecard [14]	68	68	General AI	Vague or absent documentation fails item	Unsatisfied, *unreported*
H2B	How2Bench [5]	55	55	Code	Absent documentation grouped with not done	Unsatisfied, *unreported*
ABC	ABC [28]	42	44	Agentic	Undocumented item scored 0	Unsatisfied, *unreported*
CAL	Caliper [1]	28	28	Cultural	Unknown; dimension lowered to 1–2	Unsatisfied, *unreported*
OECD	OECD [6]	18	18	All AI	Reliability allows N/A; others require value	Reliability open; others evaluated
—	**MESA Standard**	—	39	General AI	Recorded as *Undocumented* (open question)	Open

2.3 Operational Envelopes and Quality Assurance Models

Cohn and Hernández-Orallo adapted measurement review to AI benchmarks across 18 categorical facets [7,6], observing that raters agree far more often on observable operational properties (its operational envelope, such as automated scoring or environment isolation) than on evaluative quality judgments. MESA captures these properties in eight Part A descriptive criteria (A01–A08, Section 3.3) to gate which Part B evaluative checks apply. In safety-critical systems, AMLAS [12] and ML component quality models [17] structure argument patterns connecting requirements to evidence. NIST AI 800-2 [20] and Evaluation Cards [9] standardize reporting metadata, while BenchFrame [16] and BQAI [10] address benchmark repair and quality indexing. Whereas quality models specify top-down goals and BQAI aggregates ratings into a composite index, MESA reconciles and audits item-level evidence across existing review frameworks while separating absent documentation from confirmed defects.

2.4 Existing Benchmark Review Frameworks

We examine seven published frameworks evaluating AI benchmarks as measurement instruments (Table 1): **BetterBench (BB)** [23] (47 template items, 46 in the paper); the **ECBD worksheet** [19] (20 open-ended questions); the **Scorecard of AI Benchmark Quality (SC)** [14] (22 principles and 46 sub-bullets, 68 items in all); **How2Bench (H2B)** [5] (55 checklist items for code benchmarks); the **Agentic Benchmark Checklist (ABC)** [28] (44 released items, 42 in the paper); **Benchmark Caliper (CAL)** [1] (28 items on cultural and regional transfer); and the **OECD facets** [6] (18 classification facets).

3 A Unified Meta-Evaluation Standard and Rubric

3.1 Standard Terminology and Three-Tier Architecture

Table 2 defines the eleven terms and three evaluation tiers used throughout this paper. The 39 MESA criteria comprise **8 Part A descriptive criteria (A01–A08)**, which map

Table 2. Terminology and three-tier architecture of the MESA standard.

Term	Scope / Tier	Definition and Possible Values
Framework	Source instrument	One of the 7 published benchmark checklists (Table 1)
Item	Source question	One question in a framework's template (280 items in all)
Concern	Mapping unit	One of 30 shared review concerns (`C01`–`C30`)
Link	Item allocation	Mapping from an item to a concern (386 links across 280 items)
Criterion	MESA rubric	One of 39 criteria: 8 Part A (`A01`–`A08`) + 31 Part B (`B01`–`B29`, with split items)
Answer	Study 1 item	A reviewer's response to one framework item
Classification	Tier 1 (Part A)	Category on `A01`–`A08` defining the operational envelope
Rating	Tier 1 (Part B)	One of 6 levels (*Claimed*, *Documented*, *Verified*, *Contradicted*, *Undocumented*, *N/A*) + outcome
Verdict	Tier 2 (Framework Comparison)	**Satisfied**, **Unsatisfied**, or **Open**, with *unreported* qualifier ($-r$) for documentation gaps
Decision	Tier 3 (Requirement Acceptance)	Evidence gate status (**Inadmissible**, **Inquiry Required**, **Admissible**) + score decision
Session	Study execution	One run applying a framework or MESA to a frozen benchmark snapshot

to Concern C30, and **31 Part B evaluative criteria (B01–B29)**, which map to Concerns C01–C29 (one criterion per concern except Concern C14, which splits into benchmark exposure safeguards B14a and candidate-model exposure B14b, and Concern C28, which splits into grading isolation B28a and agent environment isolation B28b).

These terms structure MESA across three tiers so evidence descriptions, framework comparisons, and requirements gates never reuse labels: (1) **Tier 1 (Rubric Ratings):** assigns Part A categories and Part B evidence levels with a separate check *outcome* (*passed*, *failed*, *not performed*, *not recorded*); (2) **Tier 2 (Harmonized Verdicts):** recodes external framework scores into **Satisfied**, **Unsatisfied**, or **Open**, attaching the ***unreported*** qualifier ($-r$) whenever an unfavorable verdict rests on absent documentation; and (3) **Tier 3 (Requirements Acceptance):** maps audited evidence to gate statuses (**Inadmissible**, **Inquiry Required**, **Admissible**) before evaluating score thresholds.

3.2 Framework Selection, Mapping Procedure, and Recoding Rules

Framework selection and concern construction. We selected seven benchmark review frameworks spanning general AI (BetterBench, Scorecard, OECD), NLP (ECBD), code generation (How2Bench), agentic tool use (ABC), and cultural transfer (Caliper). We transcribed all 280 items from released templates (Table 1) and grouped them bottom-up into 30 comparison concerns (`C01`–`C30`) across Part A (`C30`) and Kane's four inferential stages (`C01`–`C29`; Table 3).

Mapping reconciliation and compound item allocation. An assistant pass drafted concern links for all 280 items, which the author independently mapped and reconciled. The author's mapping matched the draft on 180 items and differed on 100, primarily regarding secondary concerns on compound items. Reconciliation kept the draft on 56, adopted the author's reading on 28, combined both on 4, and revised 12 (matching the draft on 234 of 280 items overall). Because 76 items ask compound questions—`SC-65`, for example, checks both scoring rules (`C06`) and edge cases (`C07`)—items map to each concern addressed, producing **386 total links** with zero unmapped items (Table 3). All 28 Caliper items map to cultural transfer (`C23/B23`), with 13 carrying secondary links to `C11`, `C13`, `C16`, or `C26`.

Recoding and concern aggregation rules. Deterministic recoding rules map item answers to Tier 2 without altering native prompts: (1) *Score and checklist frameworks:*

Table 3. The 30 comparison concerns and derived rubric criteria allocations structured across Kane's inferential stages.

Inferential Stage	MESA Criteria Scope	BB	ECBD	SC	H2B	ABC	CAL	OECD	Total
Part A: Descriptive	A01–A08 (Concern C30: OECD Profile Facets)	0	0	0	0	0	0	10	10
Scoring (I_{score})	B06 (Scoring Rule), B07 (Harness Gaming)	4	3	12	5	21	0	1	46
Generalization (I_{gen})	B08–B12, B20 (Baselines, Stability, Power)	9	8	15	11	8	1	6	58
Extrapolation (I_{extrap})	B01, B03–B05, B13–B16, B23, B27 (Tasks, Contam.)	20	13	45	24	23	40	8	173
Decision (I_{dec})	B02, B17–B19, B21, B22, B24–B26, B28, B29	23	5	16	29	15	4	7	99
Total Links	**30 Concerns (386 Allocated Framework Links)**	**56**	**29**	**88**	**69**	**67**	**45**	**32**	**386**

BetterBench scores of 15 map to **Satisfied**, 10 to **Satisfied** (*partial*), 5 to **Unsatisfied** (*unreported*), and 0 to **Unsatisfied** (with $-r$ when unreported). Scorecard `satisfied`, How2Bench `done`, and ABC 1 map to **Satisfied**; `partially_done` maps to **Satisfied** (*partial*); negative answers map to **Unsatisfied** (with $-r$ when unreported); and `not_applicable` maps to **Open**. (2) *Worksheets and facets:* ECBD answers map to **Satisfied** when evidence is cited, **Unsatisfied** for defects, and **Open** when absent. OECD evaluative facets map to **Satisfied** or **Unsatisfied**, while Part A descriptive facets (`C30`) are compared categorically. (3) *Offline filter and aggregation:* Six items requiring live hosting or citation tracking (`BB-45`, `BB-46`, `H2B-45`, `CAL-IC-4`, `CAL-OO-4`, `CAL-OC-4`) are excluded from concern aggregation. Under strict conjunction, a framework's verdict on concern c is **Unsatisfied** if any mapped item is **Unsatisfied**, **Satisfied** if all are **Satisfied**, and **Open** otherwise.

Traced example: Concern C07 (Scoring Validation) on HumanEval. Four frameworks evaluate scoring validation (`C07`): BetterBench (`BB-12`) scores 10 for pass@k validation (→ **Satisfied**, *partial*); ECBD (`ECBD-Q17`) cites Figure 8 (→ **Satisfied**); How2Bench (`H2B-22`) marks `not_mentioned` for unreported coverage (→ **Unsatisfied**, $-r$); and Scorecard (`SC-65`–`SC-68`) marks four items `unsatisfied` for absent parser checks (→ **Unsatisfied**, $-r$). Of $\binom{4}{2} = 6$ pairs, two agree (`BB`–`ECBD` and `H2B`–`SC`) while four disagree because `H2B` and `SC` penalize unmentioned checks.

3.3 The Review Criteria, Evidence Levels, and Record Schema

The 39-criterion MESA rubric consists of two parts: (1) **Part A (8 Descriptive Criteria A01–A08):** adapted from OECD facets [6] to record the operational envelope: *Purpose* (**A01**), *Judgeability* (**A02**, scoring method: manual, automated, or mixed; gates **B07**), *Reference* (**A03**: absolute or relative; gates **B09**), *Containedness* (**A04**, environment isolation: fully contained, partial interference, or not contained), *Objectivity* (**A05**), *Autonomy* (**A06**), *Progression* (**A07**, static or developmental), and *Variation* (**A08**, task perturbation: fixed, altered, or procedural; gates **B20**). (2) **Part B (31 Evaluative Criteria B01–B29, with B14a/b and B28a/b):** rated on six evidence levels (Table 4): *Claimed* (stated in text without an inspectable artifact), *Documented* (inspectable artifact present in paper or repository), *Verified* (verification check whose coverage, findings,

Table 4. Representative evaluative criteria and their level thresholds.

Id	Evaluation Criterion	What Counts as Documented	What Counts as Verified	Applies When
B01	Ability stated (Construct)	Operational form and scope defined	Definition compared or boundaries tested	Always
B05	Error checks (reject correct fix)	Checking procedure described	Coverage, findings, and limits reported	Always
B07	Exploit checks (accept wrong fix)	Scoring code inspectable	Parser/judge validated or exploits tested	Automated/mixed A02
B12	Score uncertainty reported	Sample size and uncertainty metric	Power or significance reasoning reported	Always
B14a	Exposure safeguards	Risk acknowledged; safeguard described	Contamination check or held-out set reported	Always
B14b	Candidate model exposure	Cutoff and task dates recorded	Model check or post-cutoff tasks used	Model configuration named
B18	Replication instructions	Code, data, and instructions present	Independent replication reported	Always
B28a	Grading runner isolation	Grading environment described	Isolation of outputs verified in code	Runs candidate code
B28b	Agent environment isolation	Tool/agent environment described	State clearing and isolation verified	Agent has tools or state

and limits are reported or executed), *Contradicted* (claim contradicted by an artifact at a cited location), *Undocumented* (sources do not allow a level to be chosen), and *Not applicable* (concern does not arise for a stated reason).

Why level and check outcome are recorded separately. In MESA, the rating *level* records what kind of evidence exists, whereas the check *outcome* (*passed*, *failed*, *not performed*, or *not recorded*) records what a verification check found. Consequently, a rating at level *Verified* can carry an outcome of *failed*: for example, if a reviewer executes an isolation check on a benchmark harness and proves that candidate code can overwrite the test runner, the check itself was carried out with documented scope and detection limits (level *Verified*), and its result was negative (outcome *failed*). By contrast, *Contradicted* records that an artifact directly contradicts a claim or requirement in the benchmark (such as HumanEval's execution script running unisolated `exec()` despite the paper's gVisor description). Recording the check outcome alongside the level ensures that both a *Contradicted* rating and a *Verified* rating with outcome *failed* trigger an **Inadmissible** status at a cited location (Section 6).

Besides the evidence level and check outcome, every Part B rating records five structured fields: (1) *check kind* (reported by authors, run by reviewer, both, or none); (2) *scope* (release version, named model configuration, single task, or task subset); (3) *detection limits* (known blind spots, e.g., static inspection without dynamic sandbox breakout); (4) *task defects* (individual flawed tasks cited under B05 or B07); and (5) *raters and signer* (evaluator identity, model configuration, and prompt).

The rubric specifies criterion-level rules for three split areas: (1) **B05 and B07 (defects)**, where exploit checks reach *Verified* on **B07** when every attempt is recorded alongside an independent check rejecting the exploit while accepting reference solutions; (2) **B14a and B14b (exposure)**, where **B14a** evaluates benchmark safeguards against pretraining exposure and **B14b** evaluates exposure for a named candidate model configuration; and (3) **B28a and B28b (isolation)**, where **B28a** covers grading harness isolation and **B28b** covers agent execution environment isolation from reference answers and network leakage.

4 Empirical Research Design

Following the ACM SIGSOFT Empirical Standard for Standards and Taxonomies, we evaluate MESA across three research questions: (1) **RQ1 (Framework Conflicts):** When applied to the same benchmarks, how often do existing review frameworks agree, and what causes their conflicts? (2) **RQ2 (Taxonomic Unification):** How completely can framework questions be unified into a standard that separates absent documentation from located software defects? (3) **RQ3 (Rubric Consistency):** How consistently does the unified standard evaluate benchmarks across repeated reviews, and can its ratings be verified against repository code?

4.1 Study 1: Comparing Frameworks Across Four Benchmarks

In Study 1, 18 evaluation sessions applied six of the seven mapped frameworks across four frozen benchmark snapshots: HumanEval [21] (Python function synthesis), TruthfulQA [18] (question answering), Humanity's Last Exam (HLE) [22] (multi-disciplinary academic questions), and τ-bench [26] (tool-use customer-service agents). Specifically, BetterBench, ECBD, Scorecard, and the OECD framework were applied to all four benchmarks (16 sessions), How2Bench was applied to HumanEval (1 session), ABC was applied to τ-bench (1 session), and Benchmark Caliper was not used in the 18 sessions. Two additional repeat sessions applied BetterBench and Scorecard a second time to HumanEval to measure within-framework stability. Every session ran over the frozen repository snapshot and paper text using Claude Sonnet 5.5 through Claude Code at medium effort with only two read-only file listing and file reading tools and no code execution.

4.2 Study 2: Repeatability and Source Audit

In Study 2, eight independent evaluation sessions applied the 39-criterion MESA rubric (Section 3.3) twice to each of the four frozen benchmarks using the same model, effort, read-only tools, and isolation as Study 1. The rubric, prompts, runner, and analysis plan were frozen after one pilot on a synthetic two-task benchmark and before the eight main sessions. Because the Study 2 prompt named no candidate model configuration, Criterion **B14b** was *Not applicable* in all eight sessions, and because sessions had read-only file access without code execution, reviewer checks were static repository inspections rather than executed **B07** exploit runs.

Three automated roles participated in the pipeline: (1) the *reviewer model* (Claude Sonnet 5.5 through Claude Code at medium effort), which executed all 18 Study 1 sessions, the 2 repeat sessions, and the 8 Study 2 sessions; (2) the *disagreement coder* (a Claude Fable 5.1 subagent), which classified the primary cause of each of the 143 pairwise framework disagreements in Study 1; and (3) the *source checker* (a Claude Fable 5.1 subagent), which audited all 64 Part A classifications and 248 Part B ratings in Study 2 against the cited quotes and file paths in the frozen benchmark snapshots.

5 Empirical Results

5.1 RQ1: Coverage and Disagreements Across Existing Frameworks

Answer to RQ1: *Applied to identical benchmark snapshots, existing review frameworks agreed on 159 of 302 pairwise concern comparisons (52.6%), with 92 of the 143 disagreements (64%) coded as resulting from item granularity (50) and missing-documentation rules (42).*

No single concern is covered by all seven frameworks (Table 3). In Study 1, a *pairwise comparison* consists of one benchmark b, one shared evaluative concern $c \in \{\texttt{C01}, \ldots, \texttt{C29}\}$, and two frameworks F_1, F_2 that both yield a Satisfied or Unsatisfied verdict on (b, c) after recoding. Across the 18 sessions (711 item answers), the six applied frameworks formed 302 pairwise comparisons: **159 agreed and 143 disagreed (52.6% agreement)**. Agreement was high on documentation and licensing (`B18`, `B19`: 17 of 19 pairs, 89%), but dropped on construct, scoring, and generalization criteria (`B01`–`B16`: 117 of 221 pairs, 53%).

Across framework pairs, mapped items relate in three ways: (1) *equivalent items* asking the same question with similar thresholds; (2) *overlapping items* addressing the same concern with different stringency; and (3) *grouped sub-items*, where one framework asks several narrow questions under a concern while another asks one broad question. For example, on statistical uncertainty (`C12`/`B12`), BetterBench asks whether scores report variability or confidence intervals (1 item), whereas Scorecard checks three distinct sub-items covering sample-size adequacy, error bars, and significance testing.

Coding of the 143 pairwise disagreements by the Claude Fable 5.1 disagreement coder classified **92 of 143 disagreements (64%) as arising from item granularity (50) or missing-documentation rules (42)**, compared with 42 from differing readings of ambiguous text, 8 from evidence thresholds, and 1 from source scope. Under strict conjunction, a framework with many mapped items fails a concern as soon as one narrow sub-item is unmet; in 50 of the 143 disagreements, the negative verdict was decided by a sub-item that the other framework does not ask. Under the majority-rule sensitivity check (with ties broken to Unsatisfied), overall agreement across the 302 comparisons is **148 of 302 (49.0%)**. In the two same-framework repeat sessions on HumanEval, BetterBench agreed with its first run on 19 of 21 concerns (90%) and Scorecard agreed on 20 of 22 concerns (91%).

5.2 RQ2: Separating Unreported Information from Contradicted Evidence

Answer to RQ2: *All 280 items across the seven frameworks map onto the 30 shared concerns with zero unmapped items, and MESA's unreported qualifier ($-r$) separates the 210 item answers that penalize absent reporting from the 13 answer instances that record contradictions with benchmark artifacts.*

Across the 711 item answers in Study 1, frameworks scored an absence of documentation as failing on 199 answers under strict absence-of-information rules (41 in BetterBench, 123 in Scorecard, 23 in How2Bench, and 12 in ABC). Including BetterBench answers scored 5 for lacking proof beyond a bare acknowledgment, **210 of the 711 item answers (30%)** carried the *unreported* qualifier ($-r$). By contrast, only **13 of**

the 711 item answers (1.8%) across six benchmark-concern cells recorded contradicted evidence citing a defect in the paper or repository (9 in Scorecard, 2 in BetterBench, and 2 in ABC).

An answer carrying the *unreported* qualifier ($-r$) can reflect either (1) an unmet reporting requirement, where documentation silence is itself the gap the checklist item targets, or (2) an unverified practice (such as a contamination check or parser test) that was either not performed or performed without being reported. By attaching $-r$ in Tier 2 and recording *Undocumented* (or check outcome *not performed*) separately from *Contradicted* and check outcome *failed* in Tier 1, MESA ensures that the record distinguishes absent reporting from a located defect.

5.3 RQ3: Rubric Repeatability and Source Audit

Answer to RQ3: *Across two independent runs on each of the four benchmarks, MESA matched on 26 of 32 Part A classifications (81%, $\alpha_n = 0.80$), 99 of 124 Part B evidence levels (80%, $\alpha_o = 0.70$), and 83 of 124 level-and-outcome pairs (67%), while the source audit confirmed 55 of 64 Part A and 159 of 248 Part B ratings directly against the frozen sources with 0 fabricated quotes.*

All eight Study 2 sessions completed on their first attempt at a total API cost of 2.23 USD, producing 64 Part A classifications and 248 Part B ratings. Among the 28 Part B ratings resting on absent information, 21 were recorded as *Undocumented*, 5 as *Claimed*, 2 as *Documented*, and 0 as *Contradicted*. All 35 *Not applicable* ratings stated an explicit reason, and both *Contradicted* ratings cited a repository file and line number.

Across the two runs per benchmark: (1) **Part A descriptive classifications** matched on **26 of 32 pairs (81%)** (23 exact text matches), yielding nominal Krippendorff's $\alpha_n = 0.80$; and (2) **Part B evidence levels and fields** matched on **99 of 124 pairs (80%)** (17 of 18 *Not applicable* pairs agreed). Treating the five evaluative levels as an ordinal scale across the 106 applicable pairs yields $\alpha_o = 0.70$. For structured record fields, scope matched on **124 of 124 pairs (100%)**, check kind on **101 of 124 (81%)**, check outcome on **94 of 124 (76%)**, and **evidence level and check outcome together matched on 83 of 124 pairs (67%)**.

In the source audit conducted by the Claude Fable 5.1 source checker across all 312 judgments, every cited quote was located in the frozen sources (246 of 286 matched verbatim or after whitespace normalization, and the remaining 40 spanned line breaks or ligatures in the PDF extraction). On Part A, **55 of 64 classifications (86%)** were directly supported and **9 (14%)** had weaker support (3 on `A04`, 2 each on `A07` and `A08`, and 1 each on `A03` and `A05`). On Part B, the 248 ratings fell into five audit categories: **159 (64%) directly supported**, **54 (22%) plausible absences**, **29 (12%) weaker support** (including 11 of 46 *Verified* ratings where the cited quote documented a procedure without proving verification coverage), **6 (2%) absences contradicted by the sources** (where the reviewer recorded *Undocumented* but the frozen repository or paper contained the information: a package version string in HumanEval on two ratings, sampling parameters in the HumanEval paper, a tool-enabled HLE evaluation script on two ratings, and named external reviewers in the TruthfulQA paper), and **0 unsupported** (no rating fabricated a nonexistent quote or file).

6 Discussion: Connecting Benchmark Reviews to Requirements Engineering

6.1 Aligning Requirements Validation with AI Velocity

Rapid releases of foundation models pressure engineering teams to compress requirements validation into leaderboard lookups or single-number automated benchmark audits. Our empirical findings show why this compression is fragile: published review checklists disagree on 47% of shared comparisons largely because they conflate unreported documentation with negative test outcomes (210 answers with $-r$ vs. 13 contradiction answers). In requirements engineering, however, an unverified gap in documentation and a located flaw in an evaluation harness call for different actions. Separating evidence level from check outcome allows automated tooling to inspect repositories and flag missing evidence at AI speed, while preserving human requirements engineering judgment for what cannot be automated: deciding which missing checks require vendor or maintainer inquiry (**Inquiry Required**), which verified defects block acceptance (**Inadmissible**), and what statistical threshold θ a model must clear.

6.2 A Requirements Traceability Matrix for Benchmark Evidence

To show how MESA ratings connect to downstream requirements decisions, we formulate a conceptual Requirements Traceability Matrix (RTM) design (an executable implementation and empirical case evaluation of this design on coding agents is presented in companion work). The RTM is a four-tuple relation:

$$\mathcal{M}_{\text{RTM}} \subseteq \mathcal{R} \times \mathcal{C} \times \mathcal{T} \times \mathcal{E}$$

linking a system functional requirement $r \in \mathcal{R}$, construct validity criteria C (`B01`–`B04`), a benchmark task subset $T_r \subseteq \mathcal{T}$, and audited Part B evidence records $\mathcal{E}$ for a candidate model configuration k_r, minimum task count $n_{\min}(r)$, target threshold θ, and confidence level $1 - \alpha$.

Step 1: Criterion status. Before any model score is checked, each criterion c in the requirement's profile $\tau_C(r)$ is evaluated. In plain terms: (i) if c is legitimately *Not applicable* for r ($\text{Active}(c) = 0$), its status is **N/A**; (ii) if c has a located defect ($\text{Def}(c) \equiv \text{level}(c) = \text{Contradicted} \vee \text{out}(c) =$ failed with a cited location $\text{Loc}(c)$), its status is **Inadmissible**; (iii) if c has no defect, its scope covers the retained tasks and configuration ($\text{Cov}(c, r) \equiv T'_r \cup \{k_r\} \subseteq \text{scope}(c)$), and it meets the requirement's sufficiency rule $\text{Suff}(c, r)$ (such as $\text{level}(c) \geq$ Documented for documentation criteria, or $\text{level}(c) \geq$ Verified with $\text{out}(c) =$ passed and stated detection limits for checking criteria `B07`, `B14a`, `B14b`, `B28a`, `B28b`), its status is **Admissible**; and (iv) otherwise its status is **Inquiry Required**:

$$v(c, r) = \begin{cases} \textbf{N/A} & \text{if } \text{Active}(c) = 0, \\ \textbf{Inadmissible} & \text{if } \text{Active}(c) = 1 \wedge \text{Def}(c) \wedge \text{Loc}(c), \\ \textbf{Admissible} & \text{if } \text{Active}(c) = 1 \wedge \neg\text{Def}(c) \wedge \text{Cov}(c, r) \wedge \text{Suff}(c, r), \\ \textbf{Inquiry Required} & \text{otherwise.} \end{cases} \tag{2}$$

Table 5. Instantiated requirements traceability matrix entry from the two Study 2 sessions on HumanEval.

System Req (r)	**Construct Validity** (C)	**Benchmark Tasks** ($\mathcal{T}$)	**Audited MESA Evidence** ($\mathcal{E}$)	**Status and Decision**
R1: IDE Python synthesis, pass@1 $\geq$ 80%	Function-level Python synthesis (B01: *Documented* in both runs)	HumanEval: 164 functions (B03: *Documented* in both)	B07: *Documented*, code inspected, no exploit check; B14b: N/A in session (no model named), applies to R1 without rating; B28a: *Contradicted* (run 1) / *Documented* (run 2), outcome *failed* in both at harness line 50	Status **Inadmissible** in both runs (B28a); no score decision issued. B07 and B14b flag inquiry.

Individual tasks with located test defects $D(c)$ under `B05` or `B07` are removed prior to scoring ($T'_r = T_r \setminus \bigcup_{c \in \tau_C(r)} D(c)$), so localized task bugs do not invalidate an entire benchmark when the remaining task set T'_r passes.

Step 2: Requirement evidence status. Across the requirement's profile $\tau_C(r)$, a single **Inadmissible** criterion makes the evidence **Inadmissible**; otherwise, any **Inquiry Required** criterion, fewer than $n_{\min}(r)$ retained tasks, or an empty active profile makes the status **Inquiry Required** (generating the query set $Q(r) = \{c \in \tau_C(r) \mid v(c, r) = \textbf{Inquiry Required}\}$); otherwise the evidence is **Admissible**:

$$\text{Status}(r) = \begin{cases} \textbf{Inadmissible} & \text{if } \exists c \in \tau_C(r) : v(c, r) = \textbf{Inadmissible}, \\ \textbf{Inquiry Required} & \text{else if } \exists c \in \tau_C(r) : v(c, r) = \textbf{Inquiry Required}, \text{ or } |T'_r| < n_{\min}(r), \text{ or } \tau_{\text{active}}(r) = \emptyset, \\ \textbf{Admissible} & \text{otherwise.} \end{cases} \tag{3}$$

Step 3: Score decision. Only when $\text{Status}(r) = \textbf{Admissible}$ does the team evaluate the model's score on T'_r using the $(1 - \alpha)$ confidence interval $[\text{LB}_{1-\alpha}, \text{UB}_{1-\alpha}]$ against threshold θ:

$$\text{Decision}(M, r, \theta, \alpha) = \begin{cases} \textbf{Accept} & \text{if } \text{LB}_{1-\alpha}(\text{Score}(M, T'_r)) \geq \theta, \\ \textbf{Reject} & \text{if } \text{UB}_{1-\alpha}(\text{Score}(M, T'_r)) < \theta, \\ \textbf{Inconclusive} & \text{otherwise (the interval contains } \theta\text{; reported with } n = |T'_r|). \end{cases} \tag{4}$$

Table 5 returns to our HumanEval running example for a requirement R1 (pass@1 $\geq$ 80% on docstring-to-code synthesis) with profile $\tau_C(\text{R1}) = \{\texttt{B01}, \texttt{B03}, \texttt{B07}, \texttt{B14b}, \texttt{B28a}\}$. In Study 2, the two repetitions assigned different evidence levels to `B28a` (*Contradicted* in repetition 1 vs. *Documented* in repetition 2), but both recorded check outcome *failed* at line 50 of the execution harness because `exec()` runs untrusted code in the test namespace without the gVisor sandbox described in the paper. Because Equation 2 checks $\text{Def}(c) \equiv \text{level}(c) = \text{Contradicted} \vee \text{out}(c) = \text{failed}$, both repetitions produce $v(\texttt{B28a}, \text{R1}) = \textbf{Inadmissible}$ and withhold the score decision, while flagging `B07` (untested scorer) and `B14b` (unassessed candidate-model exposure) for inquiry.

7 Limitations

Mapping construction and unreported qualifier. The 30 comparison concerns and recoding rules are a taxonomy reconciled by one author with an assistant draft (matching on 234 of 280 items after resolving 100 initial differences), not an inter-rater study among multiple human experts. Compound items map across multiple concerns, and the *unreported* qualifier ($-r$) pools unmet reporting requirements with unverified practices. The 13 contradicted answers measure log instances rather than unique bugs.

Model reviews and same-family subagents. All review sessions were executed by Claude Sonnet 5.5 through Claude Code at medium effort with read-only file access; disagreement coding and source audits used Claude Fable 5.1 subagents without human raters. Reported agreement rates ($\alpha_n = 0.80$ on Part A, $\alpha_o = 0.70$ on Part B levels, 67% on level plus outcome) measure algorithmic repeatability under a fixed configuration rather than human inter-rater reliability. Read-only sessions without named candidate models did not execute dynamic `B07` exploit tests or `B14b` model exposure checks.

Benchmark scope and practitioner decisions. The empirical evaluation covers four frozen Python benchmark snapshots (HumanEval, TruthfulQA, HLE, and τ-bench). Evaluating human reviewer effort, inter-rater agreement among requirements engineers, and the effect of the RTM design on practitioner procurement decisions remains future work.

8 Conclusion

MESA unifies seven AI benchmark review frameworks by mapping 280 items across 386 links onto 30 shared concerns and deriving a 39-criterion rubric separating evidence levels from check outcomes. Across 18 review sessions on four benchmarks, published frameworks agreed on 53% of pairwise comparisons, with 64% of conflicts stemming from item granularity and missing-documentation rules, and 210 answers penalizing unreported information compared with 13 contradiction instances. Across two runs per benchmark, MESA achieved 81% Part A and 80% Part B repeatability, with 0 fabricated quotes in independent source auditing.

Data Availability

In accordance with the REFSQ 2027 Open Science Policy and double-blind review requirements, the complete replication package is available in an anonymized repository provided with this submission, containing the mapping, recoding rules, rubric, four benchmark snapshots, 26 study session logs, and deterministic verification scripts.

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